



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PRE - SCORE

08-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	B,C,D	A,B,C	B,C	B,C	C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.49 to 2.50	0.50	166.66 to 166.67	6.92 to 6.93	0.25	7.00	330.00	8.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,B,C	A,D	A,B,D	B,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	6.25	17.66 or 17.67	7.60	0.60	471.00	4.58	13.30	5.12

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,C,D	A,B,C	A,C	A,C	B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	2.00	7921.00	4.50	43.00	142.00	112.00	11.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C,D)

$$H = -K_0 x 4\pi x^2 \frac{dT}{dx}$$

$$\frac{H}{4\pi K_0} \int_{R_0}^{2R_0} \frac{dx}{x^3} = - \int_{2T_0}^{T_0} dT, \text{ it will give } H$$

$$\text{and } \frac{H}{4\pi K_0} \int_{R_0}^x \frac{dx}{x^3} = - \int_{2T_0}^T dT, \text{ it will give } T = f(x).$$

2. Ans (B,C,D)

$$\text{Optical path length} = \int_0^D \mu dy = 2\mu_0 D$$

$$t = \frac{2\mu_0 D}{c}$$

PART-1 : PHYSICS

SECTION-I (ii)

4. **Ans (B,C)**

$$\frac{N_0}{2} = N_0 - kt^2$$

$$kt^2 = \frac{N_0}{2}$$

$$t = \sqrt{\frac{N_0}{2k}} \text{ here } t \text{ is the time at which nuclei}$$

become is half

$$N = N_0 - kt^2$$

$$dN = -k(2t)dt$$

$$\frac{-dN}{dt} = 2k\sqrt{\frac{N_0}{2k}} = \sqrt{2kN_0}$$

$$t_{av} = \frac{\int_0^{\sqrt{\frac{N_0}{k}}} (-k2tdt)t}{N_0}$$

$$t_{av} = \frac{2}{3} \sqrt{\frac{N_0}{k}}$$

5. **Ans (B,C)**

Force on the pine board F = rate of change of momentum

$$= \frac{10 - 1}{0.002} \text{ N}$$

$$= 4500 \text{ N}$$

Stress in the board due to this force is

$$\sigma = \frac{F}{A} = \frac{4500}{9.0 \times 10^{-4}} \text{ N/m}^2$$

$$= 5.0 \times 10^6 \text{ N/m}^2$$

The board will break since stress of $5.0 \times 10^6 \text{ N/m}^2$ is more than the breaking stress of $3.6 \times 10^6 \text{ N/m}^2$.

7. **Ans (A)**

In figure (A)

$$\frac{1}{f_e} = (1.5 - 1) \left(\frac{1}{\infty} - \frac{1}{-20} \right) = +\frac{1}{40}$$

$$\frac{1}{f_m} = \frac{1}{10}$$

$$\frac{1}{f'} = \frac{1}{40} + \frac{1}{10} + \frac{1}{40} = \frac{3}{20}$$

$$f' = \frac{20}{3} \text{ cm (concave mirror)}$$

$$\text{Power} = +15 \text{ D}$$

$$\text{Velocity of image} = 2 \text{ cm/s}$$

In figure (B)

Focal length of device = 4 cm and it is a

concave mirror of power = +25 D

$$\text{Velocity of image} = 4 \text{ cm/s}$$

In figure (C)

Focal length = 40 cm and it is a concave mirror

of power = +2.5 D

$$\text{Velocity of image} = 0.5 \text{ cm/s}$$

In figure (D)

Focal length = 2.5 cm (concave mirror)

$$\text{Power} = +40 \text{ D}$$

$$\text{Velocity of image} = 4 \text{ cm/s}$$

8. Ans (B)

(A)

Process is isobaric

$$\text{So } C_V = \frac{5}{2}R, C_P = \frac{7}{2}R$$

$$\Delta Q = \frac{7}{2}RT_0, \Delta U = \frac{5}{2}RT_0$$

$$\Delta W = RT_0$$

(B) $\frac{P}{T} = \text{constant}$, isochoric process

$$C = C_V = \frac{3}{2}R$$

$$\Delta Q = \Delta U = \frac{3}{2}RT_0$$

$$\Delta W = 0$$

$$(C) C_V = \frac{R}{1.5 - 1} = 2R$$

$VT = \text{constant}$; $PV^2 = \text{constant}$; $\frac{P}{\rho^2} = \text{constant}$

$$V + T \frac{dV}{dT} = 0$$

$$\frac{dV}{dT} = -\frac{V}{T}$$

$$\frac{P}{n} \frac{dV}{dT} = -\frac{PV}{nT} = -R$$

$$C = C_V - R = R$$

$$\Delta Q = RT_0, \Delta U = 2RT_0; \Delta W = -RT_0$$

$$(D) \gamma = 1 + \frac{2}{4} = \frac{3}{2}$$

$$C_V = \frac{R}{\frac{3}{2} - 1} = 2R$$

$PT = \text{constant}$; $P^2V = \text{constant}$; $\frac{P^2}{\rho} = \text{constant}$

$$\Delta U = 2RT_0$$

$$\Delta W = 2RT_0$$

$$\Delta Q = 4RT_0$$

9. Ans (D)

C.M. of the cube comes down through a height of $\sqrt{3}m$ while C.M. of displaced water goes up

by a height of $\frac{\sqrt{3}}{2}m$.

$$\Rightarrow \text{Loss in PE of cube} = 2000 \times 10 \times \sqrt{3} \text{ Joule.}$$

$$= 2\sqrt{3} \times 10^4 \text{ Joule.}$$

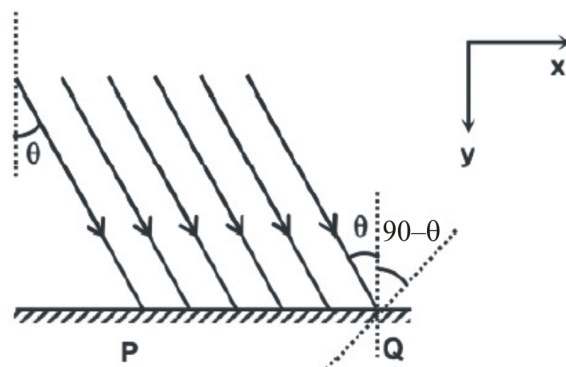
Gain in PE of water

$$= 1000 \times 10 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2} \times 10^4 \text{ Joule}$$

Using work energy theorem :

$$W_r + W_{\text{Gravity}} = 0$$

10. Ans (D)



$$\text{Intensity is given by } I = \frac{N \left(\frac{hc}{\lambda} \right)}{(A \cos \theta) \Delta t},$$

where $N \rightarrow$ total number of photons and

$\lambda \rightarrow$ wavelength of photons.

$$(A) F_x = \frac{\Delta P_x}{\Delta t} = \frac{IA \cos \theta \cdot \sin \theta}{C}$$

$$F_y = \frac{\Delta P_y}{\Delta t} = \frac{IA \cos \theta \cos \theta}{C}$$

$$p_r = \frac{F_y}{A} = \frac{I \cos^2 \theta}{C}$$

$$(B) F_x = 0$$

$$F_y = \frac{2IA \cos^2 \theta}{C} \quad p_r = \frac{2I \cos^2 \theta}{C}$$

(C) and (D)

$$F_x = \frac{IA \cos \theta \sin \theta}{C} (1 - r)$$

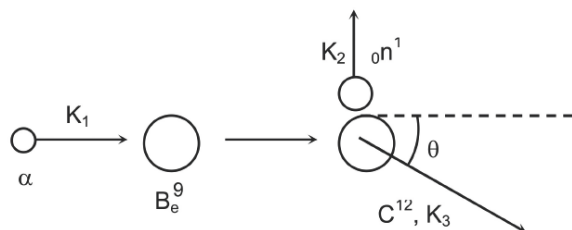
$$F_y = \frac{IA \cos^2 \theta}{C} (1 + r)$$

$$p_r = \frac{I \cos^2 \theta}{C} (1 + r)$$

PART-1 : PHYSICS

SECTION-II

1. Ans (2.49 to 2.50)



Conservation of momentum

$$\sqrt{2 \times 4K_1} = \sqrt{2 \times 12K_3} \cos \theta$$

$$\text{and } \sqrt{2 \times 1 \times K_2} = \sqrt{2 \times 12K_3} \sin \theta$$

$$\text{and } Q = K_3 + K_2 - K_1$$

$$\Rightarrow 5.80 = K_2 + K_3 - 5.32$$

Solving it $K_3 = 2.49 \text{ MeV}$.

2. Ans (0.50)

At a point at depth y from free water surface, pressure is $P = ypg$. Due to this pressure, water molecules are compressed. Hence, an elastic deformation energy (strain energy) is stored in water molecules. That strain energy per unit volume is given by $\frac{P^2}{2B}$.

Now consider a thin layer of water of thickness dy at depth y from free water surface

Volume of this layer = $A dy$

$\therefore$ Strain energy stored in water molecules of this layer,

$$dU = \frac{P^2}{2B} A dy = \frac{\rho^2 g^2 A}{2B} y^2 dy$$

$\therefore$ Total strain energy stored in water of the vessel,

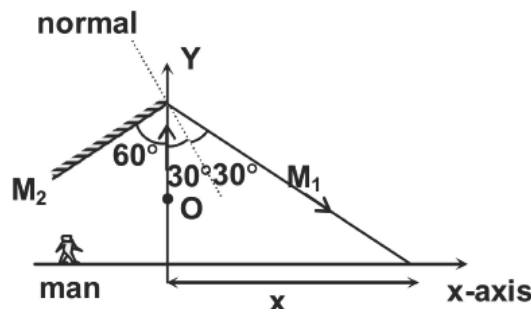
$$U = \int dU = \frac{\rho^2 g^2 A}{2B} \int_0^h y^2 dy$$

$$\text{or } U = \frac{\rho^2 g^2 A h^3}{6B} = 5 \times 10^{-7} \text{ J}$$

3. Ans (166.66 to 166.67)

$$\frac{10^5 \times 10^{-4} (1300 - 300) \times 5}{300} = 166.66$$

4. Ans (6.92 to 6.93)



$$\tan 60^\circ = \frac{x}{2}$$

$$x = 2\sqrt{3}m$$

Hence total length = $4\sqrt{3}m$.

5. Ans (0.25)

$$\text{Power, } P = mgv_0$$

$$= 2 \times 10 \times 2.5$$

$$= 50 \text{ W}$$

$$P = 2H \frac{dT}{dt}$$

$$\frac{dT}{dt} = \frac{50}{200} = 0.25^\circ \text{C/s}$$

6. Ans (7.00)

$$p - p_0 = \frac{4S}{r} \Rightarrow dp = \frac{4S}{r^2} dr$$

$$C = C_v + \frac{dW}{ndT} = C_v + \left[R + \frac{16S\pi}{3} \frac{rdr}{ndT} \right] = \frac{7R}{2}$$

7. Ans (330.00)

$$\lambda_0 - V_s t = x_1$$

$$\lambda_0 - V_s t = x_2$$

$$V_s = \frac{x_2 - x_1}{2T} = 60 \text{ m/s} \text{ \& } \lambda_0 = \frac{x_1 + x_2}{2} = 1.1 \text{ m}$$

$$\text{Speed of sound } V = n \times \lambda_0 = 330 \text{ m/s}$$

Frequency detected at point Q

$$n = \frac{V}{V - V_s} \times 300 \text{ Hz} = 330 \text{ Hz}$$

8. Ans (8.00)

$$E_{n \rightarrow m} : E_{m \rightarrow 2} : E_{2 \rightarrow 1} = 5 : 72 : 243$$

$E_{m \rightarrow 2} \Rightarrow$ Transition corresponding to visible region.

$$E_{n \rightarrow 2} : E_{2 \rightarrow 1} = 77 : 243 \Rightarrow \left(\frac{1}{2^2} - \frac{1}{n^2} \right) :$$

$$\Rightarrow \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = 77 : 243 \Rightarrow n = 9.$$

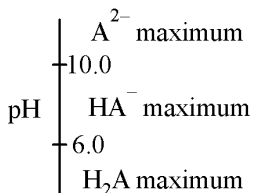
PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B)**

The orbital is $3p_z$ orbital

2. **Ans (A,B,C)**



3. **Ans (A,D)**

Theory based

$$s\text{-orbital} : A(\theta, \phi) = \sqrt{\frac{1}{4\pi}}$$

4. **Ans (A,B,D)**

(A) Reaction is endothermic.

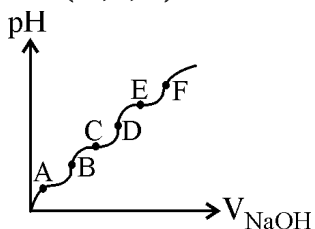
(B), (C) on adding water, equilibrium shifts right but no change in concentrations.

(D) Reaction shifts left.

5. **Ans (B,C)**

Reaction is exothermic.

6. **Ans (A,B,C)**



A,C,E $\Rightarrow$ Buffer

B,D,E $\Rightarrow$ Equivalent point

$$\text{pH at D} = \frac{1}{2} (pK_{a2} + pK_{a3}) = 9.50$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (B)**

$$(P) \frac{mV^2}{r} = K \cdot \frac{Ze^2}{r^2}$$

$$(Q) mvr = \frac{nh}{2\pi} \text{ and } r \propto n^2$$

$$(R) \frac{1}{2} mV^2 = K \cdot \frac{Ze^2}{2r}$$

$$(S) \frac{mV^2}{r} = K \cdot \frac{Ze^2}{r^2}$$

8. **Ans (A)**

$$(P) [H^+] = \sqrt{K_{a1} \times 0.1} = 10^{-4} M$$

$$(Q) \text{pH} = pK_{a1}$$

$$(R) \text{pH} = \frac{1}{2} (pK_{a1} + pK_{a2})$$

$$(S) \text{pH} = pK_{a2}$$

9. **Ans (B)**

Increase in volume will result decrease in concentrations of all components except reaction (Q).

10. **Ans (C)**

$$\frac{1}{\lambda_P} = R_H \times 1^2 \times \left(\frac{1}{1^2} - \frac{1}{4^2} \right) = \frac{15R_H}{16}$$

$$\frac{1}{\lambda_Q} = R_H \times 1^2 \times \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R_H}{36}$$

$$\frac{1}{\lambda_R} = R_H \times 1^2 \times \left(\frac{1}{3^2} - \frac{1}{5^2} \right) = \frac{16R_H}{225}$$

$$\frac{1}{\lambda_S} = R_H \times 1^2 \times \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3R_H}{4}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.25)**

$$\Delta E = h\nu$$

$$\frac{5.5 \times 10^6}{6 \times 10^{23}} \times \left(\frac{1}{4^2} - \frac{1}{5^2} \right) = 6.6 \times 10^{-34} \times \nu$$

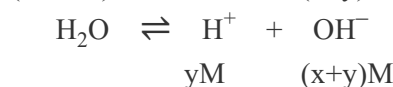
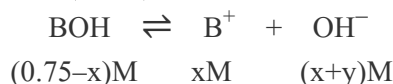
$$\therefore \nu = 3.125 \times 10^{14} \text{ Hz.}$$

2. **Ans (17.66 or 17.67)**

$$\text{For most probable distance, } \frac{d}{dr} [4\pi r^2 \psi^2(r)] = 0$$

$$\therefore r = \frac{a_0}{Z} = \frac{0.53 \times 10^{-10}}{3} = 17.67 \times 10^{-12} \text{ m.}$$

3. **Ans (7.60)**

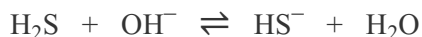


$$2 \times 10^{-13} = \frac{x \cdot (x+y)}{(0.75-x)} \approx \frac{x \cdot (x+y)}{0.75}$$

$$\text{and } 1 \times 10^{-14} = y \cdot (x+y)$$

$$\therefore \text{pH} = -\log(y) = 7.60$$

4. Ans (0.60)



Initial moles 0.2 - 0.1

NaOH added 0.1

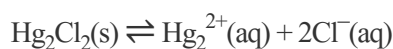
Final moles 0.1 0.2

$$\Delta\text{pH} = \text{pH}_2 - \text{pH}_1$$

$$= \left(\text{p}^{K_{a1}} + \log \frac{[\text{HS}^-]_0}{[\text{H}_2\text{S}]_0} \right)_2 - \left(\text{p}^{K_{a1}} + \log \frac{[\text{HS}^-]_0}{[\text{H}_2\text{S}]_0} \right)_1$$

$$= \log \frac{0.2}{0.1} - \log \frac{0.1}{0.2} = 0.60$$

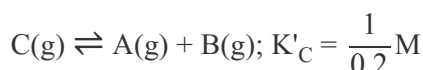
5. Ans (471.00)



$$K_{\text{sp}} = 4s^3 \Rightarrow s = \left(\frac{4 \times 10^{-18}}{4} \right)^{1/3} = 10^{-6} \text{ M}$$

$$\therefore \text{Minimum mass needed} = (10^{-6} \times 1000) \times 471 \text{ gm} = 471 \text{ mg.}$$

6. Ans (4.58)



Moles at equilibrium 5-x x x

$$\frac{1}{0.2} = \frac{x \cdot x}{5-x} \times \frac{1}{10} \Rightarrow x = 4.58$$

7. Ans (13.30)



(1-x)M xM xM

$$\frac{1}{20} = \frac{x \cdot x}{1-x} \Rightarrow x = 0.2 \Rightarrow \text{pH} = 13.30$$

8. Ans (5.12)

$$K_P = \frac{P_{\text{NO}_2}^4 \cdot P_{\text{O}_2}}{P_{\text{N}_2\text{O}_5}^2} = \frac{(0.8)^4 \times 0.2}{(0.4)^2} = 0.512 \text{ atm}^3.$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,D)

Divided the given equation by ax^4 , and let $\frac{y}{x} = m$

Equation become

$$m^4 + \frac{b}{a}m^3 + \frac{c}{a}m^2 + \frac{d}{a}m + 1 = 0 \quad \dots(1)$$

Assuming that four slopes

$$m_1, m_2, m_3, m_4 \Rightarrow m_1 \cdot m_2 \cdot m_3 \cdot m_4 = 1$$

Now it is given pair of these lines, bisect the angle

between the other pair. So let, $m_1 m_2 = -1$

$m_3 m_4 = -1$ which means other two lines are also perpendicular

Equation (1) can be written as

$$(m - m_1)(m - m_2)(m - m_3)(m - m_4) = (m - m_1)$$

$$\left(m + \frac{1}{m_1} \right) (m - m_3) \left(m + \frac{1}{m_3} \right)$$

$$= (m^2 + \lambda m - 1)(m^2 + \mu m - 1) \quad \dots(2)$$

$$\text{Where } \lambda = \frac{1 - m_1^2}{m_1} \Rightarrow \mu = \frac{1 - m_3^2}{m_3}$$

Comparing the coefficient (1) and (2)

$$\lambda + \mu = \frac{b}{a}, \lambda\mu - 2 = \frac{c}{a}, \lambda + \mu = -\frac{d}{a} \quad \dots(3)$$

$$\text{Thus, } \frac{b}{a} = -\frac{d}{a} \Rightarrow b + d = 0$$

Also, we have to use angle bisector fact

$$\left| \frac{m_1 + m_3}{1 + m_1 m_3} \right| = \left| \frac{m_3 + \frac{1}{m_1}}{1 - \frac{m_3}{m_1}} \right|$$

$$\Rightarrow (m_1 - m_3)^2 = (1 + m_1 m_3)^2 \quad \dots(4)$$

From (3),

$$\lambda\mu - 2 = \frac{c}{a} \Rightarrow \left(\frac{1 - m_1^2}{m_1} \right) \left(\frac{1 - m_3^2}{m_3} \right) - 2 = \frac{c}{a} \quad \dots(5)$$

Using (4) and (5) we obtain $(c + 6a) = 0$

2. Ans (A,C,D)

$$S_1: x^2 + y^2 - 4x - 81 = 0$$

$$S_2: x^2 + y^2 + 24x - 81 = 0$$

$$S_1 - S_2 = 0 \quad 28x = 0$$

$x = 0$ equation of AB

Now let $A \equiv (0, 9) \therefore A$ is on $x = 0$

Now put A in S_1 or S_2

$$0 + a^2 + 0 - 81 = 0$$

Now PA, $y - 9 = mx \Rightarrow y = mx + 9$

$\Rightarrow BQ, y + 9 = mx \Rightarrow y = mx - 9$

Now let $P(x_1, y_1)$ and $Q(x_2, y_2)$

Now solve S_1 and PA then

$$x_1 = \frac{-18m - 24}{m^2 + 1} \text{ and } y_1 = \frac{-9m^2 - 24m + 9}{m^2 + 1}$$

Now solve S_2 and BQ then

$$x_2 = \frac{18m + 4}{m^2 + 1} \text{ and } y_2 = \frac{9m^2 + 4m - 9}{m^2 + 1}$$

Let mid pt of PQ is (h, k) then $x_1 + x_2 = 2h$ and

$$y_1 + y_2 = 2k$$

$$x_1 + x_2 = \frac{-20}{m^2 + 1} = 2h$$

$$\frac{h}{k} = \frac{1}{m}$$

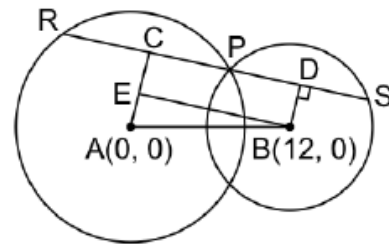
$$y_1 + y_2 = \frac{-20m}{m^2 + 1} = 2k \Rightarrow \frac{-20}{\left(\frac{k}{h}\right)^2 + 1} = 2h$$

$$h(h^2 + k^2 + 10h) = 0$$

$$E_1 \equiv x^2 + y^2 + 10x = 0$$

$$E_2 \equiv x^2 + y^2 + 6x - y + 5 = 0$$

3. Ans (A,B,C)



Let $PR = 2a$, $AC = b$, $BD = c$

$$\Delta AEB \Rightarrow 144 = (b - c)^2 + 4a^2 \dots (1)$$

$$\Delta ACP \Rightarrow 64 = a^2 + b^2 \dots (2)$$

$$\Delta PBD \Rightarrow 36 = a^2 + c^2 \dots (3)$$

$$144 = 6^2 + c^2 - 2bc + 4a^2$$

$$144 = 100 + 2a^2 - 2bc$$

$$22 = a^2 - \sqrt{64 - a^2} \cdot \sqrt{36 - a^2}$$

$$(22 - a^2)^2 = (64 - a^2)(36 - a^2)$$

$$(22 - x)^2 = 164 - x(36 - x)$$

$$484 + x^2 - 44x = 2304 - 100x + x^2$$

$$56x = 2304 - 484$$

$$56x = 1820$$

$$x = \frac{1820}{56} = 32.5$$

$$a = \sqrt{x} = \sqrt{32.5}$$

$$2a = \sqrt{4 \times 32.5} = \sqrt{130}$$

4. Ans (A,C)

$$F_1: (2x - y - 9) + \lambda(3x + y - 21) = 0$$

$$F_2: (2x + y - 5) + \mu(4x - y - 1) = 0$$

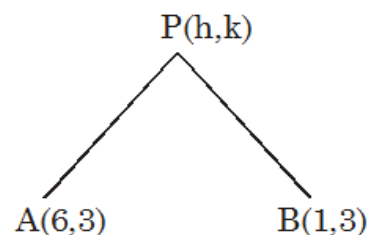
F_2 passes through $B(1, 3)$.

(A) Maximum distance between parallel line of F_1 and F_2 is distance between their fix point

$$= \sqrt{0 + 5^2} = 5$$

(C) If $L_1 \perp L_2$, then $m_{PB} \cdot m_{PA} = -1$

$$\Rightarrow \left(\frac{k-3}{h-1}\right) \left(\frac{k-3}{h-6}\right) = -1$$



$$(h-1)(h-6) + (k-3)^2 = 0$$

$$\Rightarrow (x-1)(x-6) + (y-3)^2 = 0$$

5. Ans (A,C)

$$3\sin^2 x - \cos x - 3\sin x \cos x + \sin x = 0$$

$$\text{Solving we get } \sin x = \frac{-1}{3} \text{ or } \sin x = \cos x$$

$$\text{i.e, } \sin x = \frac{-1}{3} \text{ or } \tan x = 1$$

$$A = \left\{ \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \dots, \frac{33\pi}{4} \right\}$$

6. Ans (B)

Number of possible values of a = 60, for b = pq,

If p = 3, q = 3, 5, 7, 11, 13, 17, 19

If p = 5 q = 5, 7, 11

If p = 7 q = 7

Total cases = 60 × 11 = 660

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

$$(I) \sin x = \frac{1}{2} [e^x + e^{-x}] \geq 0$$

only possibility is $\sin x = 1$ which is also not

possible because $e^x + e^{-x} = 2$ at $x = 0$

$$(II) x + y = \frac{2\pi}{3} \text{ and } \cos x + \cos y = \frac{3}{2}$$

$$2 \cos \left(\frac{x+y}{2} \right) + \cos \left(\frac{x-y}{2} \right) = \frac{3}{2}$$

$$2 \cos \frac{\pi}{3} + \cos \left(\frac{x-y}{2} \right) = \frac{3}{2}$$

$$\Rightarrow \cos \left(\frac{x-y}{2} \right) = \frac{3}{2} \text{ (Not possible)}$$

$$(III) \cos x + 2\sin x = 1 \therefore 2 \sin x = 2\sin^2 \frac{x}{2}$$

$$\sin \frac{x}{2} \left[\sin \frac{x}{2} - 2 \cos \frac{x}{2} \right] = 0$$

$$\therefore \sin \frac{x}{2} = 0, \tan \frac{x}{2} = 2$$

$$x = 2\pi \text{ and } \tan \frac{x}{2} = 2 \text{ for two values of } x$$

$$(IV) |\tan x| = \sin 2x \text{ at } x = \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$$

8. Ans (A)

$$(I) (x + y + \lambda_1)(x + y + \lambda_2) = 0$$

$$x^2 + y^2 + 2xy + x(\lambda_1 + \lambda_2) + y(\lambda_1 + \lambda_2) + \lambda_1 \lambda_2 = 0$$

$$\lambda_1 + \lambda_2 = -8\lambda, \lambda_1 \lambda_2 = -9\lambda^2$$

$$25\sqrt{2} = \left| \frac{\lambda_1 - \lambda_2}{\sqrt{2}} \right| = 50 = \sqrt{64\lambda^2 - 4(-9\lambda^2)} = 10|\lambda|$$

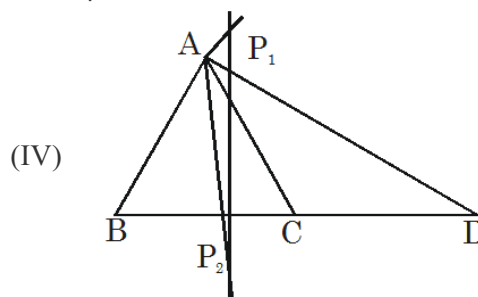
$$\Rightarrow |\lambda| = 5$$

$$(II) \lambda(\lambda - 3) + \lambda + 1 = 0$$

$$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow \lambda = 1$$

$$(III) A(1,2) \rightarrow B(2,1) \rightarrow C(2,-1) \rightarrow D(1,-2)$$

$$\alpha + \beta = 1$$



P set of points lies on perpendicular bisector of BC

Q set of point lies on angle bisector

$$n(P \cap Q) = 2$$

9. Ans (B)

$$S : x^2 + y^2 - 2x - 4y + 1 = 0,$$

centre (1, 2), radius = 2

$$S' : x^2 + y^2 - 4x - 2y + 1 = 0,$$

centre (2, 1), radius = 2

(I) (1, 3) lies inside S and outside S'.

$\Rightarrow$ Sum of tangents is 2.

(II) Required circle is

$$x^2 + y^2 - 2x - 4y + 1 + \lambda(2x - 2y) = 0$$

$$x^2 + y^2 - 2(1 - \lambda)x - 2(2 + \lambda)y + 1 = 0$$

centre is $(1 - \lambda, 2 + \lambda)$

it lies on $2x + y - 6 = 0$

$$\Rightarrow 2 - 2\lambda + 2 + \lambda - 6 = 0$$

$$-\lambda - 2 = 0 \Rightarrow \lambda = -2$$

$$\text{radius} = \sqrt{9 + 0 - 1} = \sqrt{8}$$

$$\frac{r^2}{4} = 2$$

(III) $m = -1$

(IV) Circles are intersecting

$\Rightarrow$ length of tangent

$$= \sqrt{(C_1 C_2)^2 - (r_1 - r_2)^2} = 2$$

4. **Ans (4.50)**

$\lambda(x - 2y + 3) + \mu(x + y - 6) = 0$ the point of intersection of above lines $x - 2y + 3 = 0$ and $x + y - 6 = 0$ is (3,3) which is centre of circle
Number of family of lines which cuts off equal intercept on co-ordinate axes is

$$(\lambda + \mu)x - (2\lambda - \mu)y + 3\lambda - 6\mu = 0$$

$$\frac{\lambda + \mu}{2\lambda - \mu} = -1 \Rightarrow \lambda + \mu = -2\lambda + \mu$$

$$\therefore \lambda = 0$$

Hence line is $x + y - 6 = 0$

Thus circle with centre (3,3) and cuts the line $x + y - 6 = 0$ on the co-ordinate axes is

$$(x - 3)^2 + (y - 3)^2 = (3\sqrt{2})^2$$

Hence radius is $3\sqrt{2}$ and area is 18π

5. **Ans (43.00)**

No. of observations: - 50

main $(\bar{x}) = 15$

Standard deviation (σ) = 2

Let incorrect observation is x_1 & correct observation is (x'_1)

Given $x_1 + x'_1 = 70$

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_{50}}{50} = 15 \text{ (given)}$$

$$\Rightarrow x_1 + x_2 + \dots + x_{50} = 750 \text{ ... (i)}$$

Now

Mean of correct observation is 16

$$\frac{x'_1 + x_2 + \dots + x_{50}}{50} = 16$$

$$x'_1 + x_2 + x_3 + \dots + x_{50} = 16 \times 50 \text{ ... (ii)}$$

eq. (ii) - eq. (i)

$$\Rightarrow x'_1 - x_1 = 16 \times 50 - 15 \times 50$$

$$x'_1 - x_1 = 50 \text{ \& } x_1 + x'_1 = 70$$

$$x'_1 = 60$$

$$x_1 = 10$$

$$\Rightarrow 4 = \frac{x_1^2 + x_2^2 + \dots + x_{50}^2}{50} - 15^2 \text{ ... (iii)}$$

$$\Rightarrow \sigma^2 = \frac{x_1'^2 + x_2^2 + \dots + x_{50}^2}{50} - 16^2 \text{ ... (iv)}$$

from (iii)

$$\Rightarrow 4 = \frac{(10)^2}{50} + \frac{x_2^2 + x_3^2 + \dots + x_{50}^2}{50} - 225$$

$$\Rightarrow 4 = 2 - 255 + \frac{(x_2^2 + x_3^2 + \dots + x_{50}^2)}{50}$$

$$\Rightarrow 227 = \frac{(x_2^2 + x_3^2 + \dots + x_{50}^2)}{50} \text{ from (iv)}$$

$$\sigma^2 = \frac{(60)^2}{50} + \left(\frac{x_2^2 + x_3^2 + \dots + x_{50}^2}{50} \right) - (16)^2$$

$$\sigma^2 = \frac{60 \times 60}{50} + 227 - 256$$

$$\sigma^2 = 72 + 227 - 256$$

$$\sigma^2 = 43$$

6. **Ans (142.00)**

$$\sum x_0^1 = \frac{3(1 - (\frac{1}{2}))^{20}}{1 - \frac{1}{2}} = 6 \left(1 - \frac{1}{2^{20}} \right)$$

$$= \sum_{i=1}^{20} (x_{i-i})^2$$

$$= \sum_{i=1}^{20} (x_i)^2 + (i)^2 - 2x_i i$$

Now

$$= \sum_{i=1}^{20} (x_i)^2 = \frac{9(1 - (\frac{1}{4}))^{20}}{1 - \frac{1}{4}} = 12 \left(1 - \frac{1}{2^{40}} \right)$$

$$\sum_{i=1}^{20} i^2 = \frac{1}{6} \times 20 \times 21 \times 41 = 2870$$

$$\sum_{i=1}^{20} x_i i = s = 3 + 2.3 \frac{1}{2} + 3.3 \frac{1}{2^2} + 4.3 \frac{1}{2^3} + \dots \text{AGP}$$

$$= 6 \left(2 - \frac{22}{2^{20}} \right)$$

$$\bar{x} = \frac{12 - \frac{12}{2^{40}} + 2870 - 12 \left(2 - \frac{22}{2^{20}} \right)}{20}$$

$$\bar{x} = \frac{2858}{20} + \left(\frac{-12}{2^{40}} + \frac{22}{2^{20}} \right) \times \frac{1}{20}$$

$$[\bar{x}] = 142$$

7. **Ans (112.00)**

$A = \{1, 2, 3, 4, 5, 6, 7\}$ and

$B = \{3, 6, 7, 9\}$

Total subset of $A = 2^7 = 128$

$C \cap B = \phi$ when set C contains the

element 1, 2, 4, 5

$\therefore S = \{C \subseteq A; C \cap B \neq \phi\}$

$= \text{Total} - (C \cap B = \phi)$

$= 128 - 2^4 = 112$

8. **Ans (11.00)**

$S = \{4, 6, 9\}$ $T = \{9, 10, 11, \dots, 1000\}$

$A = \{a_1 + a_2 + \dots + a_k : k \in \mathbb{N}\} \text{ \& } a_i \in S$

Here by the definition of set 'A'

$A = \{a : a = 4x + 6y + 9z\}$

Except the element 11, every element of set T is of

the form $4x + 6y + 9z$ for some $x, y, z \in \mathbb{W}$

$\therefore T - A = \{11\}$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PRE - SCORE

08-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	B	D	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C	B,D	B,C	A,D	A,B,D	A,B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	7	2	4	2	3	5	3	3

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B	A,C	B,C,D	A,B,D	B,C	A,B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	3	2	1	4	6	5	6

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	D	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C,D	D	A,B,C	A,B,C,D	B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	4	0	1	2	5	0	2	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

Least count of screw gauge = $\frac{0.5}{50} = 0.01\text{mm} = \Delta r$

Diameter $r = 2.1\text{ mm} + 20 \times \frac{0.5}{50} = 2.30\text{ mm}$

$$\frac{\Delta r}{r} = \frac{0.01}{2.30} \text{ or } \frac{\Delta r}{r} \times 100 = \frac{1}{2.30}$$

$$\text{Now density } d = \frac{m}{v} = \frac{m}{\frac{4}{3}\pi\left(\frac{r}{2}\right)^3}$$

Here, r is the diameter

$$\therefore \frac{\Delta d}{d} \times 100 = \left\{ \frac{\Delta m}{m} + 3 \left(\frac{\Delta r}{r} \right) \right\} \times 100$$

$$= \frac{\Delta m}{m} \times 100 + 3 \times \left(\frac{\Delta r}{r} \right) \times 100$$

$$= 1\% + 3 \times \frac{1}{2.30} \approx 2.4\%$$

3. Ans (D)

$$\frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{280}{320}} = \sqrt{\frac{7}{8}}$$

For Snell's law, $\mu_1 \sin i = \mu_2 \sin r$

$$\Rightarrow r = \sin^{-1} \left[\frac{\mu_1}{\mu_2} \sin i \right] = \sin^{-1} \left[\sqrt{\frac{8}{7}} \cdot \frac{1}{2} \right]$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (A,D)

x-z plane

Normal is $\hat{j}$

$$\cos A = b, \cos B = \beta \Rightarrow \sin A = \alpha, \sin B = \alpha$$

$$\mu_1 \sin A = \mu_2 \sin B \Rightarrow \mu_1 \alpha = \mu_2 \alpha$$

y-z plane

$$\text{Normal is } \hat{i}; \cos A = a, \cos B = \alpha \Rightarrow \sin A = b,$$

$$\sin B = b$$

$$\mu_1 \sin A = \mu_2 \sin B \Rightarrow \mu_1 b = \mu_2 B$$

9. **Ans (A,B,D)**

$$\rho(z) = \frac{\rho_0}{2} \left(1 + \frac{P(z)}{P_0} \right)$$

$$P(z) = \int \rho(z) g dz + P_{\text{atm}} = \int_0^z \frac{\rho_0}{2} \left(1 + \frac{P(z)}{P_0} \right) g dz + P_{\text{atm}}$$

$$\frac{dP(z)}{dz} = \frac{\rho_0}{2} \left(1 + \frac{P(z)}{P_0} \right) g$$

$$\log \left(\frac{P_0 + P(z)}{P_0 + P_{\text{atm}}} \right) = \frac{\rho_0 g z}{2P_0}$$

$$\rho(z) = \rho_0 (P_0 + P_{\text{atm}}) e^{\rho_0 g z / 2P_0}$$

$$\rho(z) = \frac{\rho_0 (P_0 + P_{\text{atm}})}{2P_0} e^{\rho_0 g z / 2P_0}$$

Mass of the liquid M

$$= \int_0^h \rho(z) A dz = \frac{A}{g} (P_0 + P_{\text{atm}}) (e^{\rho_0 g h / 2P_0} - 1)$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (7)**

$$\ell_1 = 0.516 \text{ m}, \ell_2 = 0.491 \text{ m}, T = 20 \text{ N}.$$

Mass per unit length, $\mu = 0.001 \text{ kg/m}$.

$$\text{Frequency, } v = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$v_1 = \frac{1}{2 \times 0.516} \sqrt{\frac{20}{0.001}}$$

$$v_2 = \frac{1}{2 \times 0.491} \sqrt{\frac{20}{0.001}}$$

$$\therefore \text{Number of beats} = v_1 - v_2 = 7$$

2. **Ans (2)**

For lens B according to the figure

$$u = +3$$

$$v = -6$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow f = -2 \text{ m}$$

5. **Ans (3)**

Since the mass of an electron is negligible compared to the mass of a nucleus, a β -decay process causes no appreciable change in the atomic weight of an element. In other words, the radioactive element and the stable product are isobars, and have the same atomic mass number. Therefore, instead of mass, we can analyse the number of nuclei.

Let the initial number of nuclei of X be N_0

$$\text{Then, at } t = T \text{ (before removal), } N = \frac{N_0}{2}$$

$$\text{At } t = T \text{ (after removal), } N = \frac{N_0}{4}$$

Ans so on, until at $t = 4T$ (before removal), the number of nuclei remaining is $\frac{N_0}{2} = \frac{N_0}{128}$

Therefore, the mass of the element X in the vessel at the end is $\frac{m}{128}$.

And, the total amount of material in the vessel at the end (after a total of three removals) is $\frac{m}{2^3} = \frac{m}{8}$

So, the mass of the element Y in the vessel at the end is $\frac{m}{8} = \frac{m}{128} = \frac{15}{128} m$

6. **Ans (5)**

$$E_n - E_1 = \Delta E_1 + \Delta E_2$$

$$\therefore \left[-\frac{13.6}{n^2} + 13.6 \right] (Z)^2 = \frac{12375}{1085} + \frac{12375}{304}$$

Putting $Z = 2$ for He^+ , we get $n = 5$

7. **Ans (3)**

Because the first process is isobaric, the pressure at the beginning of the isothermal expansion is P_0 . Because $PV = \text{constant}$ for the isothermal expansion, the volume at the beginning of the isothermal expansion (and hence the volume at the end of the isobaric expansion) must be $\frac{3W_0}{2}$

$$\text{Therefore, } W_1 = P \Delta V = P_0 \left(\frac{3V_0}{2} - V_0 \right) = \frac{P_0 V_0}{2}$$

Now, if the temperature at the beginning of the isothermal expansion is T_1 , we know that

$$RT_1 = P_0 \left(\frac{3V_0}{2} \right)$$

$$\text{So, } W_2 = (1) RT_1 \ln(2) = \frac{3P_0 V_0}{2} \ln 2$$

8. **Ans (3)**

Path difference at C.

$$\Delta x = t_1(\mu - 1) - t_2(\mu - 1) = \mu(t_1 - t_2) - (t_1 - t_2)$$

$$= (t_1 - t_2)(\mu - 1) = (2.5 - 1.25) \left(\frac{1.4 \times 3}{4 \times 10} - 1 \right)$$

$$= 1.25 \times \frac{2}{40} = \frac{2.5}{400} \Rightarrow \Delta x = \frac{1}{16} \mu\text{m}$$

$$\phi = \frac{2\pi}{\lambda} \cdot \Delta x = \frac{2\pi \times 4}{5000 \times 3 \times 10^{-10}} \cdot \frac{1}{16} \times 10^{-16}$$

$$\Rightarrow I_{\text{max}} = 4I_0$$

$$I \text{ at C. } I_c = 2I_0(1 + \cos \frac{\pi}{3}) = 3I_0 ;$$

$$\text{Required ratio } \frac{I_c}{I_0} = 3$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**



$$K_{\text{eq}} = \frac{1}{K_a} = \frac{K_b}{K_w} = 5 \times 10^4$$

$$\begin{array}{ccc} 0.5\text{M} & 0.2\text{M} & 0 \end{array}$$

$$\text{Final } \begin{array}{ccc} 0.3\text{M} & 0 & 0.2\text{M} \end{array}$$

$$\text{pH} = \text{pKa} + \log \frac{[\text{CH}_3\text{COO}^-]_0}{[\text{CH}_3\text{COOH}]_0}$$

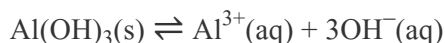
$$= 4.70 + \log \frac{0.3}{0.2} = 4.88$$

2. **Ans (A)**

$$(\text{K.E.})_{\text{max}} = \frac{1240}{206.67} - 4 = 2 \text{ eV}$$

$$\therefore \lambda_{\text{min}} = \sqrt{\frac{150}{2}} = 8.66 \text{ \AA}$$

3. **Ans (B)**



$$\frac{1}{2} = 0.5\text{M}$$

$$[\text{OH}^-]_{\text{max}} = \left(\frac{4 \times 10^{-18}}{0.5} \right)^{1/3} = 2 \times 10^{-6} \text{ M}$$

$$\therefore \text{pOH}_{\text{min}} = 5.70 \text{ and } \text{pH}_{\text{max}} = 8.30$$

4. **Ans (C)**

$$p_{\text{H}_2\text{O}} = \sqrt{K_P} = 0.03 \text{ atm.}$$

To drive the reaction in backward direction,

$$p_{\text{H}_2\text{O}} > 0.03 \text{ atm}$$

$$\text{or, } \left(0.04 \times \frac{x}{100} \right) > 0.03 \Rightarrow x > 75$$

$$\therefore \text{Min. R.H.} = 75\%.$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B)**

$$\text{For } K_P < K_C, \Delta n_g = -\text{ve.}$$

6. **Ans (A,C)**

$$Q = \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}} = \frac{(2.0)^2}{0.2} = 20 \text{ atm} < K_P$$

Reaction will occur in forward direction.

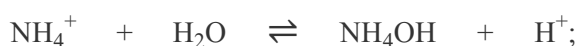
Density will not change, as mass remain constant.

7. **Ans (B,C,D)**

$$\text{Radial node} = 4 - 2 - 1 = 1 \text{ (spherical)}$$

$$\text{Angular node} = \ell = 2 \text{ (nodal cones)}$$

8. **Ans (A,B,D)**



$$K_h = \frac{K_w}{K_b} = \frac{10^{-9}}{1.8}$$

$$\begin{array}{ccc} (0.5-x)\text{M} & x\text{M} & x\text{M} \end{array}$$

$$\frac{10^{-9}}{1.8} = \frac{x^2}{(0.5-x)} \approx \frac{x^2}{0.5} \Rightarrow x = \frac{10^{-4}}{6}$$

$$h = \frac{x}{0.5} = \frac{10^{-4}}{3}$$

$$\text{pH} = -\log \left(\frac{10^{-4}}{6} \right) = 4.78$$

$$n_{\text{NH}_4\text{OH}} = \frac{x \times 100}{1000} = \frac{10^{-5}}{6}$$

9. **Ans (B,C)**

$$[\text{Cl}^-]_{\min} \text{ needed to start precipitation of AgCl} \\ = \frac{2 \times 10^{-10}}{0.5} = 4 \times 10^{-10} \text{ M.}$$

$$[\text{Cl}^-]_{\min} \text{ needed to start precipitation of Hg}_2\text{Cl}_2 \\ = \sqrt{\frac{8 \times 10^{-17}}{0.2}} = 2 \times 10^{-8} \text{ M} > 4 \times 10^{-10} \text{ M}$$

Hence, Ag^+ will start first.

$$\text{Now, } [\text{Ag}^+] \text{ when precipitation of Hg}_2\text{Cl}_2 \text{ just start} \\ = \frac{2 \times 10^{-10}}{2 \times 10^{-8}} = 0.01 \text{ M.}$$

$$\therefore \% \text{ Ag}^+ \text{ precipitated} = \frac{0.5 - 0.01}{0.5} \times 100 \\ = 98\%.$$

10. **Ans (A,B,D)**

$$\text{A(s)} \rightleftharpoons 2\text{C(g)} + \text{D(g)}; K_{P_1} = (40)^2 \times 20 \\ = 32000 \text{ torr}^3$$

$$\text{B(s)} \rightleftharpoons 2\text{C(g)} + \text{E(g)}; K_{P_2} = (60)^2 \times 30 \\ = 108000 \text{ torr}^3$$

$$\frac{n_{\text{A(g)}}}{n_{\text{E(g)}}} = \frac{K_{P_1}}{K_{P_2}} = \frac{8}{27}$$

$$P_{\text{total}} = P_{\text{C(g)}} + P_{\text{D(g)}} + P_{\text{E(g)}} \\ = (2x + 2y) + x + y = 3x + 3y$$

$$\text{And, } (2x + 2y)^2 \cdot x = 32000$$

$$(2x + 2y)^2 \cdot y = 108000$$

$$\therefore x + y = (35000)^{1/3} = 32.71$$

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (3)**

$$[\text{H}^+] = \sqrt{K_a \cdot C} = \sqrt{2 \times 10^{-5} \times 0.05} = 10^{-3} \text{ M} \\ \therefore \text{pH} = 3.0$$

2. **Ans (3)**

$$r_n = \frac{0.529 n^2}{Z} \text{ \AA} \\ \Rightarrow 2.82 = \frac{0.529 \times 4^2}{Z} \\ \Rightarrow Z = 3$$

3. **Ans (2)**

$$\begin{array}{ccc} \text{A} & \rightleftharpoons & 2\text{B} \\ \text{Initial mole} & a & 0 \\ \text{Equilibrium} & a-x & 2x \\ \text{Equ. conc.} & \frac{a-x}{10} \text{ M} & \frac{2x}{10} \text{ M} = 0.2 \text{ M} \end{array}$$

$$\therefore x = 1$$

$$\text{Now, } K_C = \frac{[\text{B}]^2}{[\text{A}]} \Rightarrow 0.4 = \frac{(0.2)^2}{\left(\frac{a-x}{10}\right)} \Rightarrow a = 2$$

4. **Ans (1)**

$$\text{A}_2\text{B}_3(\text{s}) \rightleftharpoons 2\text{A}^{3+}(\text{aq}) + 3\text{B}^{2-}(\text{aq}) \\ \begin{array}{ccc} 2 \text{ SM} & & 3 \text{ SM} \end{array} \\ K_{\text{sp}} = [\text{A}^{3+}]^2 [\text{B}^{2-}]^3 = (2\text{S})^2 \cdot (3\text{S})^3 = 108 \text{ S}^5$$

$$\therefore \text{S} = \left(\frac{1.08 \times 10^{-18}}{108} \right)^{1/5} = 10^{-4} \text{ M}$$

$$\text{Millimoles dissolved} = (10^{-4} \times 10) \times 10^3 = 1.0$$

5. **Ans (4)**

$$\text{S} = [\text{Cd}^{2+}] + [\text{Cd}(\text{OH})_4^{2-}] = \frac{K_1}{[\text{OH}^-]^2} + K_2 \cdot [\text{OH}^-]^2$$

$$\text{For minimum solubility, } \frac{dS}{d[\text{OH}^-]} = 0$$

$$\therefore [\text{OH}^-] = \left(\frac{K_1}{K_2} \right)^{1/4} = 10^{-4} \text{ M}$$

$$\Rightarrow \text{pOH} = 4.0$$

6. **Ans (6)**

$$\text{No. of available orbits} = 4$$

$$\therefore \text{Number of spectral lines} = {}^4\text{C}_2 = 6.$$

7. **Ans (5)**

$$\lambda = 3.32 \frac{n}{Z} \text{ \AA} \Rightarrow 8.3 = \frac{3.32 \times n}{2} \Rightarrow n = 5.$$

8. **Ans (6)**

$$\text{pH range} = 7.0 - 9.0$$

$$\text{pOH range} = 5.0 - 7.0$$

$$\therefore \text{pK}_b = \frac{5.0 + 7.0}{2} = 6.0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)

Parametric equation of line through A is

$$x = -2 + r \cos \theta$$

$$y = -3 + r \sin \theta$$

It intersect $x + 3y = 9$ and $x + y + 1 = 0$ at B and C

$$(-2 + r \cos \theta) - 9 + 3r \sin \theta = 9$$

$$r (\cos \theta + 3 \sin \theta) = 20$$

$$r = \frac{20}{\cos \theta + 3 \sin \theta} = AB$$

$$-2 + r \cos \theta + (r \sin \theta - 3) + 1 = 0$$

$$r = \frac{4}{\cos \theta + \sin \theta} = AC$$

$$AB \cdot AC = 20$$

$$\frac{20}{(\cos \theta + 3 \sin \theta)} \cdot \frac{4}{(\cos \theta + \sin \theta)} = 20$$

$$4 = (\cos \theta + \sin \theta)(\cos \theta + 3 \sin \theta)$$

$$4 = \cos^2 \theta + 4 \sin \theta \cos \theta + 3 \sin^2 \theta$$

$$8 = 1 + \cos 2\theta + 4 \sin 2\theta + 3 - 3 \cos 2\theta$$

$$4 = 4 \sin 2\theta - 2 \cos 2\theta$$

$$2 = 2 \sin 2\theta - \cos 2\theta$$

$$2 (\sin \theta - \cos \theta)^2 = -(\cos^2 \theta - \sin^2 \theta)$$

$$2 (\sin \theta - \cos \theta)^2 = \cos \theta + \sin \theta$$

$$\sin \theta = 3 \cos \theta$$

$$\tan \theta = 3$$

$$\tan \theta = 1$$

$$\therefore \begin{cases} x - y = 1 \\ 3x - y + 3 = 0 \end{cases}$$

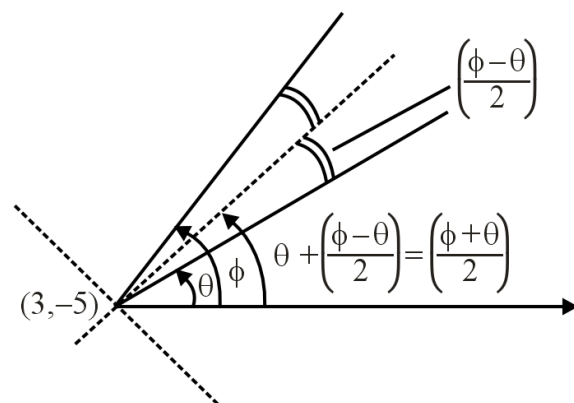
6. Ans (A,B,C,D)

$\therefore$ One bisector makes an angle $\left(\frac{\theta + \phi}{2}\right)$ with

X-axis,

then other bisector makes an angle

$90^\circ + \left(\frac{\theta + \phi}{2}\right)$ with X-axis.



$\therefore$ Equation of bisectors are

$$\frac{x - 3}{\cos\left(\frac{\theta + \phi}{2}\right)} = \frac{y + 5}{\sin\left(\frac{\theta + \phi}{2}\right)} \quad \dots (i)$$

$$\text{and } \frac{x - 3}{\cos\left(\frac{\pi}{2} + \frac{\theta + \phi}{2}\right)} = \frac{y + 5}{\sin\left(\frac{\pi}{2} + \frac{\theta + \phi}{2}\right)}$$

$$\Rightarrow \frac{x - 3}{-\sin\left(\frac{\theta + \phi}{2}\right)} = \frac{y + 5}{\cos\left(\frac{\theta + \phi}{2}\right)} \quad \dots (ii)$$

$$\text{But given bisector are } \frac{x - 3}{\cos \alpha} = \frac{y + 5}{\sin \alpha}$$

$$\therefore \alpha = \frac{\theta + \phi}{2} \text{ and } \frac{x - 3}{\beta} = \frac{y + 5}{\gamma}$$

[from Eq. (i)]... (iii)

$$\therefore \beta = -\sin\left(\frac{\theta + \phi}{2}\right) = -\sin \alpha$$

$$\text{[from eq. (ii)] And } \gamma = \cos\left(\frac{\theta + \phi}{2}\right) = \cos \alpha$$

7. Ans (D)

For R to be reflexive $\Rightarrow x R x$

$$\Rightarrow 3x + \alpha x = 7x \Rightarrow (3 + \alpha)x = 7K$$

$$\Rightarrow 3 + \alpha = 7\lambda \Rightarrow \alpha = 7\lambda - 3 = 7N + 4, K, \lambda, N \in I$$

$\therefore$ when α divided by 7, remainder is 4.

R to be symmetric $xRy \Rightarrow yRx$

$$3x + \alpha y = 7N_1, 3y + \alpha x = 7N_2$$

$$\Rightarrow (3 + \alpha)(x + y) = 7(N_1 + N_2) = 7N_3$$

Which holds when $3 + \alpha$ is multiple of 7

$$\therefore \alpha = 7N + 4 \text{ (as did earlier)}$$

R to be transitive

$$xRy \text{ \& } yRz \Rightarrow xRz$$

$$3x + \alpha y = 7N_1 \text{ \& } 3y + \alpha z = 7N_2$$

$$\text{and } 3x + \alpha z = 7N_3$$

$$\therefore 3x + 7N_2 - 3y = 7N_3$$

$$\therefore 7N_1 - \alpha y + 7N_2 - 3y = 7N_3$$

$$\therefore 7(N_1 + N_2) - (3 + \alpha)y = 7N_3$$

$$\therefore (3 + \alpha)y = 7N$$

Which is true again when $3 + \alpha$ divisible by 7,

i.e. when α divided by 7, remainder is 4.

8. Ans (A,B,C)

For intersection of S_3 and S_4 orthogonally

$$2(1)(0) + 2(k)(k) = 6 + k$$

$$2k^2 = 6 + k$$

$$2k^2 - k - 6 = 0$$

$$(k - 2)(2k + 3) = 0$$

$$\therefore k = 2, k = -\frac{3}{2}$$

Hence A and B.

Equation of circle through intersection of S_1 and S_2 is

$$S = (x^2 + y^2 - 6) + \lambda(x^2 + y^2 - 6x + 8) = 0$$

S passes through (1,1)

$$\therefore \lambda = 1$$

$$\text{Hence : } S = x^2 + y^2 - 3x + 1 = 0$$

9. Ans (A,B,C,D)

(A) Mid point of (a, b) & (3, 8) lies on $x + 3y = 7$

$$\Rightarrow \frac{a+3}{2} + \frac{3(b+8)}{2} = 7 \Rightarrow a + 3b + 27 = 14$$

$$\Rightarrow a + 3b + 13 = 0$$

$$\text{Also } m_1 m_2 = -1$$

$$\Rightarrow m_1 m_2 = -1 \Rightarrow \left(\frac{b-8}{a-3}\right) \left(-\frac{1}{3}\right) = -1$$

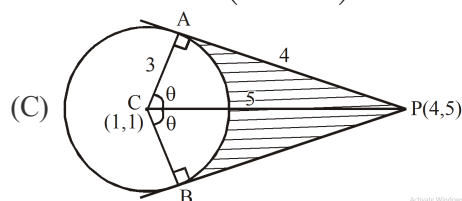
$$\Rightarrow 3a = b + 1$$

$$\Rightarrow a = -1 \text{ \& } b = -4 \Rightarrow a - b = 3$$

$$(B) \Delta = \frac{1}{2} \begin{vmatrix} \ell & \ell & 1 \\ \ell & 2\ell & 1 \\ 2\ell & \ell & 1 \end{vmatrix} = \frac{1}{2} (\ell^2 + \ell^2 - 3\ell^2)$$

$$= \frac{\ell^2}{2} = 18 \Rightarrow \ell = \pm 6$$

$$\Rightarrow \text{centroid is } \left(\frac{4\ell}{3}, \frac{4\ell}{3}\right) \Rightarrow \frac{|a| + |b|}{4} = 4$$



$$\text{Shaded area} = \text{ar(PCAB)} - \text{ar sector(ABC)} =$$

$$4 \times 3 - \frac{1}{2} \times 9 \times 2\theta$$

$$= 12 - 9\theta$$

$$= 12 - 9 \tan^{-1} \frac{4}{3} \Rightarrow \frac{k}{2} = 6$$

$$(D) x^2 y + x^2 - y^3 + 3y^2 - 4 = 0$$

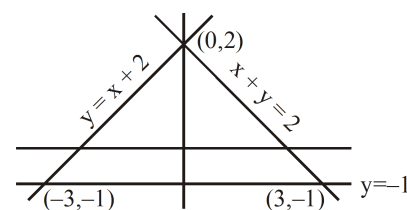
$$x^2 y + x^2 - y^3 - y^2 + 4y^2 - 4 = 0$$

$$x^2(y+1) - y^2(y+1) + 4(y+1)(y-1) = 0$$

$$\Rightarrow (x^2 - y^2 + 4y - 4)(y+1) = 0$$

$$\Rightarrow x^2 = y^2 - 4y + 4 \text{ or } y = -1$$

$$x = y - 2 \text{ or } y = -1$$



$$A = \frac{1}{2} \times 3 \times 6$$

$$A = 9 \text{ sq. units}$$

10. Ans (B,C,D)

$$\Delta = \begin{vmatrix} \sin \theta & \cos \theta & \cos \theta \\ \cos \theta & \sin \theta & \cos \theta \\ \cos \theta & \cos \theta & \sin \theta \end{vmatrix}$$

$$= (\sin \theta + 2 \cos \theta) \begin{vmatrix} 1 & 1 & 1 \\ \cos \theta & \sin \theta & \cos \theta \\ \cos \theta & \cos \theta & \sin \theta \end{vmatrix}$$

$$[R_1 \rightarrow R_1 + R_2 + R_3]$$

$$= (\sin \theta + 2 \cos \theta) \begin{vmatrix} 1 & 0 & 0 \\ \cos \theta & \sin \theta - \cos \theta & 0 \\ \cos \theta & 0 & \sin \theta - \cos \theta \end{vmatrix}$$

$$\begin{bmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{bmatrix}$$

$$= (\sin \theta - \cos \theta)^2 (\sin \theta + 2 \cos \theta)$$

Now, if $\tan \theta = -2$ then $\Delta = 0$

$$\text{But, } h^2 - ab = \cos^2 \theta - \sin^2 \theta = \cos 2\theta = -\frac{3}{5} < 0$$

i.e. imaginary pair of lines

$$\text{Hence, for real straight lines, } \theta = n\pi + \frac{\pi}{4}$$

$$\text{Also, in this case } h^2 - ab = \cos 2\theta = 0$$

i.e. lines are parallel.

$$\text{further, one can observe that by taking } \theta = \frac{\pi}{2},$$

$$S = 0 \text{ becomes } x^2 + y^2 + 1 = 0$$

i.e. no real locus.

PART-3 : MATHEMATICS

SECTION-III

1. Ans (4)

Straight line from Origin is $y = mx$

$$4x + 2y = 3, y = mx,$$

$$x_1 = \frac{3}{(2m+4)}, y_1 = \frac{3m}{(2m+4)}$$

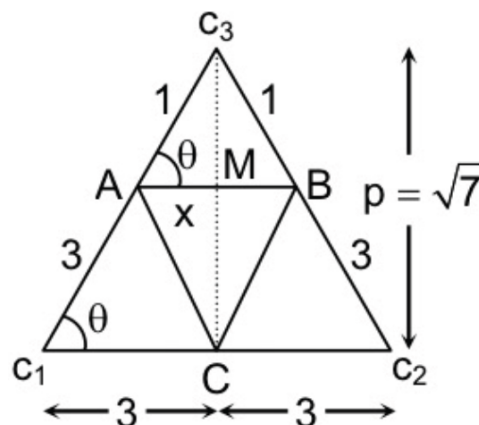
$$2x + y + 6 = 0, y = mx,$$

$$x_2 = \frac{-6}{(m+2)}, y_2 = \frac{-6m}{(m+2)}$$

$$O = \frac{\frac{-6m}{(m+2)} + \frac{3mk}{2(m+2)}}{k+1} \Rightarrow O = \frac{\frac{-6}{m+2} + \frac{3k}{2(m+2)}}{k+1}$$

$$= \frac{-6 + \frac{3k}{2}}{k+1} = 0 \Rightarrow \frac{3}{2}k = 6, k = \frac{12}{3} = 4$$

2. Ans (0)



$$\cos \theta = \frac{3}{4} = \frac{x}{1}$$

$$\Rightarrow x = \frac{3}{4} \Rightarrow AB = \frac{3}{2}$$

$$p = \sqrt{4^2 - 3^2} = \sqrt{7} \text{ and } C_3M = \frac{\sqrt{7}}{4}$$

$$\Rightarrow CM = \sqrt{7} - \frac{\sqrt{7}}{4} = \frac{3\sqrt{7}}{4}$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2}(AB)(CM)$$

$$= \frac{9}{16}\sqrt{7}$$

3. Ans (1)

$$\Rightarrow (x^2 - 1)^2 = -3\{1 + \cos(ax^2 + bx + c)\}$$

solution exists when L.H.S. = R.H. S. = 0

$$\Rightarrow x = \pm 1 \text{ and } \cos(ax^2 + bx + c) = -1$$

$$\text{When } x = 1, x = -1$$

$$\therefore a, b, c \in (2, 5) \Rightarrow a + b + c \in (6, 15) \text{ \&}$$

$$a - b + c \in (-1, 8)$$

$$\Rightarrow a + b + c = 3\pi, a - b + c = \pi$$

$$\Rightarrow b = \pi \text{ and } a + c = 2\pi$$

$$\therefore \frac{a+c}{2} \geq \sqrt{ac} \therefore ac \leq \pi^2 \therefore \frac{ac}{b^2} \leq 1$$

4. **Ans (2)**

From (i) $2\cos^2 x - 1 + 1 - \sqrt{3} = (2 - \sqrt{3}) \cos x$

$$2\cos^2 x - (2 - \sqrt{3}) \cos x - \sqrt{3} = 0$$

$$2\cos x (\cos x - 1) + \sqrt{3}(\cos x - 1) = 0$$

$$(\cos x - 1)(2\cos x + \sqrt{3}) = 0$$

$$\cos x = 1 \text{ or } \cos x = \frac{-\sqrt{3}}{2}$$

From (ii) $\sin 3x = 2 \sin x$

$$3 \sin x - 4 \sin^3 x = 2 \sin x$$

$$\sin x = 0 \text{ or } 3 - 4 \sin^2 x = 2$$

$$4 \sin^2 x = 1$$

$$\sin x = \pm \frac{1}{2}$$

common solution in $[0, 5\pi]$ is

$$0, 2\pi, 4\pi, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{17\pi}{6}, \frac{19\pi}{6}, \frac{29\pi}{6}$$

If $\tan x \geq \frac{1}{\sqrt{3}}$

$$x \in \left[\frac{\pi}{6} + n\pi, \frac{\pi}{2} + n\pi \right)$$

then common solution: $x \in \frac{7\pi}{6}, \frac{19\pi}{6}$

5. **Ans (5)**

$$2^{2\sin^2 x - 3 \sin x + 1} + 2^{3 \sin x - 2\sin^2 x + 2} = 9$$

$$\Rightarrow 2^{2\sin^2 x - 3 \sin x + 1} + 2^{2 - (2\sin^2 x - 3 \sin x)} = 9$$

$$2^{2\sin^2 x - 3 \sin x} = t$$

$$2t + \frac{4}{t} = 9 \Rightarrow t = 4 \text{ or } t = \frac{1}{2}$$

$$\Rightarrow 2^{2\sin^2 x - 3 \sin x} = 2^2 \text{ or } 2^{2\sin^2 x - 3 \sin x} = 2^{-1}$$

$$\Rightarrow 2\sin^2 x - 3 \sin x = 2$$

$$\text{or } 2\sin^2 x - 3 \sin x = -1$$

$$\Rightarrow \sin x = -\frac{1}{2}, 1, \frac{1}{2}, 2 \text{ (reject)}$$

5 solutions in $[0, 2\pi]$

6. **Ans (0)**

$$C_1(-5, 12) \quad r_1 = 16$$

$$C_2(5, 12) \quad r_2 = 4$$

Let P be the centre of the third circle and r be

its radius. then $PC_1 = 16 - r$. $PC_2 = 16 - r$. PC_2

$$= 4 + r \text{ then } PC_1 + PC_2 = 16 - r + 4 + r = 20$$

So, P is on the ellipse centred at (0, 12) with

foci C_1 and C_2 so major axis = 20 and minor

axis = $10\sqrt{3}$. so equation of ellipse

$$\frac{x^2}{100} + \frac{(y - 12)^2}{75} = 1$$

now y-axis satisfy it

$$\frac{x^2}{100} + \frac{(ax - 12)^2}{15} = 1$$

$$D \geq 0 \Rightarrow a^2 \geq \frac{69}{100}$$

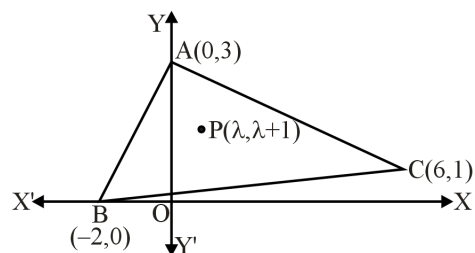
$$\Rightarrow [m] = 0$$

7. Ans (2)

Equation of AB is $3x - 2y + 6 = 0$

Equation of BC is $8 - 8y + 2 = 0$,

Equation of CA is $x + 3y - 9 = 0$



Let $P \equiv (\lambda, \lambda + 1)$ $\because$ B and P lie on one side of AC, then

$$\frac{\lambda + 3(\lambda + 1) - 9}{-2 + 0 - 9} > 0 \quad \text{or} \quad 4\lambda - 6 < 0 \quad \text{or}$$

$$\lambda < \frac{3}{2} \quad \dots (i)$$

and C and P lie on one side of AB, then

$$\frac{3\lambda - 2(\lambda + 1) + 6}{18 - 2 + 6} > 0 \quad \text{or} \quad \lambda + 4 > 0 \quad \text{or}$$

$$\lambda > -4 \quad \dots (ii)$$

Finally, A and P lie on one side of BC, then

$$\frac{\lambda - 8(\lambda + 1) + 2}{0 - 24 + 2} > 0 \quad \text{or} \quad -7\lambda - 6 < 0 \quad \text{or}$$

$$\lambda > -\frac{6}{7} \quad \dots (iii)$$

From Eqs. (i), (ii) and (iii), we get $-\frac{6}{7} < \lambda < \frac{3}{2}$

Integral value of λ are 0 and 1.

Hence, number of integral value of λ is 2.

8. Ans (4)

$$\sum x_1 = 15 \times 20 = 300 \quad \dots (i)$$

$$\frac{\sum x_1^2}{20} - (15)^2 = 9 \quad \dots (ii)$$

$$\sum x_1^2 = 234 \times 20 = 4680$$

$$\frac{\sum (x_1 + a)^2}{20} = 178 \Rightarrow \sum (x_1 + a)^2 = 3560$$

$$\Rightarrow \sum x_1^2 + 2a \sum x_1 + \sum a^2 = 3560$$

$$4680 + 600a + 20a^2 = 3560$$

$$\Rightarrow a^2 + 30a + 56 = 0$$

$$\Rightarrow (a + 28)(a + 2) = 0$$

$$a = -2, -28$$

Square of maximum value of a is 4